Gabriel Mareze

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Education

University of Central Florida — Orlando, FL
Bachelor of Science in Mechanical Engineering | GPA: 3.65 — *Expected Dec. 2027*

Experience

Astranis Space Technologies — San Francisco, CA
Mechanical Engineering Intern, Spacecraft Configuration Team — *Aug. 2026 – Present*

- Cleared flight hardware for procurement by drafting and presenting Critical Design Reviews (CDRs) to directors and senior engineers, defending design rationale and analysis results on parts I designed.
- Owned 50+ waveguide support brackets across 2 satellite configurations, designing and analyzing aluminum and PEEK brackets in Ansys FEA, then installing them and other hardware onto the spacecraft.
- Owned Reaction Wheel Test Fixtures, designing a 3D printed fixture that positions calibrated magnets at precise Gauss levels from a PCB-mounted Hall-effect sensor to find its switching threshold.
- Manage large multi-subsystem SolidWorks top-level assemblies using top-down modeling, catching hole geometry disparities and manufacturability issues before release, and releasing parts through Arena PLM.
- Wrote and released company-wide Python-based SolidWorks macros that auto-fillet large parts, standardize fillet sizing, and flag broken fillet features to cut diagnosis time and reduce CNC operations.
- Designed DEV and Flight PCB enclosures for thermal performance, adding extra surface geometry for heat dissipation along with cable management and flight-qualified hardware.

Design Launchers, LLC — Orlando, FL
Product Development Engineer Associate — *Jul. 2025 – Aug. 2026*

- Designed and managed 27 products and 250+ SolidWorks components across consumer, medical, and industrial applications, with 12 products advanced to active manufacturing.
- Led modeling and prototyping phases across multiple projects, developing designs from initial sketches through manufacturable products.
- Created and reviewed 50+ manufacturing drawings using datums, tolerances, and GD&T to improve part fit and reduce manufacturing errors.
- Developed 35+ mechanical and electromechanical prototypes using 3D printing, laser cutting, machining, and sheet metal fabrication, integrating PCBs, wiring, and soldered electronics.
- Worked with 9 manufacturing vendors across CNC machining, sheet metal, injection molding, and additive manufacturing to prototype and produce components.
- Conducted customer design reviews, trained other interns, and took ownership of projects from initial concept through production.

Projects

Autonomous Warehouse Robot Fleet - Design Launchers | *SolidWorks, Tolerance Analysis* — *2025 – 2026*

- Led a full mechanical redesign after diagnosing mid-development instability, shifting the CG to balance the frame under load across 12 redesigned and newly created components.
- Delivered 20 higher payload capacity than competing platforms and hit a sub $1,000 unit cost target through design optimization and cross-team vendor coordination.
- Created and redlined 28 manufacturing prints, catching tolerance errors and material mismatch prior to release.

NASA Student Launch: Project Zenith | *OpenRocket, Onshape, PID Control* — *Aug. 2024 – May 2025*

- Wrote a PID controller on an ESP32 to deploy and retract 4 airbrake flaps mid-flight, targeting a 4,850 ft apogee against a simulated 5,234 ft overshoot.
- Validated the system across 6 subscale flights, hitting within an average of 2.5 feet of target apogee.
- Diagnosed and solved in-flight power failure via multimeter, traced to an unsoldered battery connection.

Skills

CAD/Software: SolidWorks (1650 hrs), Ansys, MATLAB, Python, OpenRocket, Excel
Engineering: GD&T, Tolerance Stack-Up, DFM, FEA, PID Control
Fabrication: Injection Molding, CNC Machining, Turning, Sheet Metal, SLA/FDM, Rotomolding, Laser Cutting
Other: Rapid Prototyping, Material Selection, Technical Drawing, Client Communication